STAGE 5 INDUSTRIAL TECHNOLOGY ELECTIVE - METAL INDUSTRY

Sheet-metal Toolbox | 25-period teaching program

Website-complete teacher review copy | Industrial Technology 7-10 Syllabus (2025)

Course	Stage 5 Industrial Technology elective - Metal industry	Delivery	2027 | 10 weeks
Timetable	25 x 60-minute periods	Pattern	5 periods per fortnight
Lesson balance	5 theory | 5 project/folio | 15 practical	Syllabus	2025 syllabus - approved early use
Status	Local teacher review only	Project	Produce the approved sheet-metal toolbox project

Website coverage: 16 named theory sections, 7 quiz/check sets (20 questions) and 5 scaffolded responses are scheduled. Practical lessons remain the majority.

Industrial Technology outcomes

Outcome	Official wording
INT5-SAF-01	applies risk management and safe work practices in industrial technology contexts
INT5-COM-01	communicates ideas and processes in industrial technology contexts
INT5-ENV-01	assesses the impact of industrial activities and practices on society and the environment
INT5-DES-01	selects and applies design, project management and technical skills in project production
INT5-USE-01	selects and justifies the use of materials, tools and equipment to undertake projects
INT5-EVL-01	evaluates the functional, aesthetic, economic and environmental qualities of products
INT5-INP-01	analyses and applies a range of industrial processes in production contexts
INT5-PRO-01	demonstrates practical skills in the production of projects
INT5-IND-01	investigates and explains a range of industry careers, enterprise opportunities and practices

Teacher to confirm before use

When	Confirm	Owner
Before Period 1	Assessment dates, submission method, class supports and adjustments	Teacher / faculty
Before practical work	Current plan revision, permitted variations, induction, SOPs, tools, PPE, supervision and risk controls	Teacher / faculty
Before finishing	Approved finish, SDS/product instructions, area, drying and storage arrangements	Teacher / faculty

Teaching program | Periods 1-10

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
1	Site/theory	Module 1: website	Read and discuss: Risk management is a process, not a poster; Risk management is a process, not a poster knowledge check (2 questions); written response. Save or show the completed website evidence.	Risk management and risk-reduction procedures	Website evidence checked INT5-SAF-01
2	**Project/folio**	Module 1: folio	Read and complete: Safe workshop practices. Update the folio with the current decision, process evidence and next step.	Risk management and risk-reduction procedures	Folio + website evidence INT5-SAF-01
3	**Practical**	Module 1: site + practical	First 20 min: Incident response and reporting. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Risk management and risk-reduction procedures	Website + practical check INT5-SAF-01, INT5-COM-01
4	**Practical**	Module 1: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Risk management and risk-reduction procedures	Practical quality check INT5-SAF-01
5	**Practical**	Module 1: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Risk management and risk-reduction procedures	Photo + quality note INT5-SAF-01
6	Site/theory	Module 2: website	Read and discuss: Choose metal for the job; Choose metal for the job knowledge check (2 questions); written response. The Toolbox brief and production context. Save or show the completed website evidence.	Metal properties and justified material selection	Website evidence checked INT5-USE-01, INT5-ENV-01
7	**Project/folio**	Module 2: folio	Read and complete: Design and project planning. Update the folio with the current decision, process evidence and next step.	Accurately measure and mark out metal	Folio + website evidence INT5-USE-01, INT5-PRO-01
8	**Practical**	Module 2: site + practical	First 20 min: Read the Toolbox plan before marking out; Read the Toolbox plan before marking out knowledge check (10 questions). Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Accurately measure and mark out metal	Website + practical check INT5-USE-01, INT5-PRO-01
9	**Practical**	Module 2: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Accurately measure and mark out metal	Practical quality check INT5-USE-01, INT5-PRO-01
10	**Practical**	Module 2: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Accurately measure and mark out metal	Photo + quality note INT5-USE-01, INT5-PRO-01

Teaching program | Periods 11-20

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
11	Site/theory	Module 3: website	Read and discuss: Accuracy before force; Accuracy before force knowledge check (2 questions); written response. Save or show the completed website evidence.	Quality checks for metals and components	Website evidence checked INT5-EVL-01, INT5-PRO-01
12	**Project/folio**	Module 3: folio	Read and complete: Marking and measuring tools. Update the folio with the current decision, process evidence and next step.	Accurately measure and mark out metal	Folio + website evidence INT5-USE-01, INT5-PRO-01
13	**Practical**	Module 3: site + practical	First 20 min: Cutting and forming sequence. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Cut, shape and form sheet metal and metal sections	Website + practical check INT5-INP-01, INT5-PRO-01
14	**Practical**	Module 3: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Accurately measure and mark out metal	Practical quality check INT5-USE-01, INT5-PRO-01
15	**Practical**	Module 3: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Accurately measure and mark out metal	Photo + quality note INT5-USE-01, INT5-PRO-01
16	Site/theory	Module 4: website	Read and discuss: A joint is a design decision; A joint is a design decision knowledge check (2 questions); written response. Save or show the completed website evidence.	Design factors and project specifications	Website evidence checked INT5-DES-01, INT5-COM-01
17	**Project/folio**	Module 4: folio	Read and complete: Machining is permission-based. Update the folio with the current decision, process evidence and next step.	Use machining and fabrication tools and equipment	Folio + website evidence INT5-SAF-01, INT5-INP-01
18	**Practical**	Module 4: site + practical	First 20 min: Surface preparation and protection. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Prepare metal surfaces for finishes	Website + practical check INT5-INP-01, INT5-PRO-01
19	**Practical**	Module 4: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Use machining and fabrication tools and equipment	Practical quality check INT5-SAF-01, INT5-INP-01
20	**Practical**	Module 4: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Use machining and fabrication tools and equipment	Photo + quality note INT5-SAF-01, INT5-INP-01

Teaching program | Periods 21-25

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
21	Site/theory	Module 5: website	Read and discuss: Quality, technology and evaluation; Quality, technology and evaluation knowledge check (1 question); written response. Save or show the completed website evidence.	Quality checks for metals and components	Website evidence checked INT5-EVL-01, INT5-PRO-01
22	**Project/folio**	Module 5: folio	Read and complete: Current and emerging manufacturing. Update the folio with the current decision, process evidence and next step.	Document, reflect on and evaluate project development	Folio + website evidence INT5-COM-01, INT5-DES-01
23	**Practical**	Module 5: site + practical	First 20 min: Folio and final evaluation; Folio and final evaluation knowledge check (1 question). Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Quality checks for metals and components	Website + practical check INT5-EVL-01, INT5-PRO-01
24	**Practical**	Module 5: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Quality checks for metals and components	Practical quality check INT5-EVL-01, INT5-PRO-01
25	**Practical**	Module 5: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Quality checks for metals and components	Photo + quality note INT5-EVL-01, INT5-PRO-01

Evidence, feedback and practical controls

Check	Look for	Respond
Website	Named theory, checks and responses completed	Revisit feedback and correct missed ideas
Folio	Decisions and process evidence tied to the project	Set one clear next step
Practical	Safe method, accuracy, quality and clean-down	Pause, demonstrate, practise and recheck

Use the current approved plan, local induction, SOPs, PPE, supervision, risk controls and product instructions for every practical lesson.